SUMIT PAUL

Bangalore, India
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SUMMARY

Software Engineer with hands-on experience in **backend, full-stack, frontend, and AI-powered application development**. Skilled in building **REST APIs, microservices, database-backed applications, authentication systems, and responsive web applications** using Java, Python, JavaScript, TypeScript, React, Node.js, Express.js, and FastAPI. Strong foundation in **Data Structures & Algorithms, Object-Oriented Programming, DBMS, System Design, API Design, and Cyber Security**.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, JavaScript, TypeScript
Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication
Frontend & Web: React, SvelteKit, Tailwind CSS, HTML5, CSS3, Responsive Design
Databases & Storage: MongoDB, MySQL, SQLite, NoSQL, Redis
AI & Application Development: Gemini AI, Groq LLM, Llama 3.3-70b, OCR.Space
Cloud, Deployment & Tools: Firebase, Cloud Functions, Docker, Docker Compose, Git, GitHub, VS Code, MongoDB Atlas, Figma
Core Computer Science: Data Structures & Algorithms, OOP, DBMS, System Design, API Design, Computer Networks, Cyber Security
Data & Visualization: Tableau, Power BI, Matplotlib, Microsoft Excel, Chart.js

PROJECTS

FinPilot AI – Smart Expense Tracker GitHub — Sept 2025 – Dec 2025

SvelteKit, Node.js/Express, FastAPI, MongoDB, Groq LLM, OCR.Space, Docker, Redis, Tailwind CSS

- Engineered a financial platform processing **50+ receipt formats** using OCR.Space and Groq Llama 3.3-70b, achieving **95%+ accuracy**, with an AI financial advisor and offline fallback.
- Designed a **3-tier microservices architecture** with SvelteKit frontend, Express REST API with **15+ endpoints**, FastAPI microservice, MongoDB, Redis, and JWT authentication.
- Implemented bank SMS bulk importing for **15+ Indian bank formats** using regex-based extraction, supporting INR, USD, EUR, and GBP, with Docker Compose deployment and real-time financial health scoring.
- Built Chart.js analytics dashboards for monthly trends, category breakdowns, and weekly comparisons with **3D visualizations** and multi-currency financial tracking.

Community Hero – AI-Powered Civic Engagement Platform GitHub — Jun 2026 – Jul 2026

React, TypeScript, Firebase, Cloud Functions, Firestore, Gemini AI, Leaflet, Tailwind CSS

- Architected a full-stack civic reporting platform supporting **1,000+ concurrent users**, integrating Gemini AI for automatic issue categorization and **Firestore** with fine-grained security rules and **RBAC across 3+ role levels**.
- Implemented geo-tagged reporting with Leaflet, real-time analytics tracking **500+ reported issues**, and a community verification workflow for hyperlocal infrastructure reporting.
- Built a hotspot prediction engine using **Geohashing and nightly cron jobs**, alongside a citizen reputation and gamification system.

Image Steganography GitHub — Dec 2024 – Feb 2025

Java, Swing, LSB Steganography, AES-256/CBC Encryption, SHA-256 Hashing

- Engineered a **1,277-line Java Swing application** implementing LSB steganography with AES-256/CBC encryption and SHA-256 password hashing for text and audio concealment across JPG, PNG, and BMP formats.
- Achieved **99.2% embedding capacity** with zero perceptual quality loss across three image formats and validated security against LSB steganalysis attacks using statistical analysis.

Weather Forecast Website GitHub — Mar 2024 – Apr 2024

HTML5, CSS3, Vanilla JavaScript, OpenWeatherMap API, Geolocation API

- Developed a responsive weather application with HTML5 Geolocation, global city search, and **OpenWeatherMap** API integration for **40+ weather parameters**, hourly forecasts, and extended 4-day forecasts.
- Implemented mobile-first CSS Grid/Flexbox layouts with three responsive breakpoints, dynamic DOM generation, **5+ micro-interactions**, keyframe animations, and optimized load time of **under 2 seconds**.

HACKATHONS & ACHIEVEMENTS

Smart India Hackathon 2025 – Top 15 Teams, Grand Finale — 2025

- Designed system architecture and developed the Flask backend for **AyurSutra**, an AI-powered Ayurvedic patient management platform featuring automated therapy scheduling, JWT authentication, EHR integration, and n8n workflow automation using React, Python Flask, and MSSQL.
- Applied object-oriented design and agile development practices in a multi-tiered distributed system alongside a cross-functional team.

Vibe2Ship – National-Level AI Hackathon Participant — 2026

Coding Ninjas × Google for Developers

Certifications: Data Structures and Algorithms with Java (Alpha) – Apna College | Introduction to Networking and Cloud Computing – Microsoft (2024) | Oracle Certified Foundations Associate | Generative AI Professional – Oracle University

EDUCATION

New Horizon College of Engineering, Bangalore — 2023 – 2027

B.E. – Computer Science and Engineering (VTU) — **CGPA: 9.51**